Project Initialization and Planning Phase

Date	02 July 2026
Team ID	xxxxxx
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to improve the credit card approval process using Machine Learning. The system analyzes an applicant's personal and financial information to predict whether the credit card application is likely to be approved or rejected. It reduces manual effort, increases prediction accuracy, and provides instant results through a Flask-based web application.

Project Overview	
Objective	TheDevelop an intelligent machine learning system that predicts credit card approval accurately and provides users with real-time results through a simple web interface.
Scope	TThe project includes data collection, exploratory data analysis, preprocessing, feature engineering, model training, evaluation, deployment using Flask, and online prediction of credit card approval status.
Problem Statement	
Description	Banks often rely on lengthy manual verification processes to evaluate credit card applications. This increases processing time and .may lead to inconsistent approval decisions
Impact	An automated machine learning solution improves decision accuracy, reduces processing time, minimizes manual effort, and enhances customer satisfaction.
Proposed Solution	
Approach	Develop a supervised machine learning model using applicant information such as age, income, employment status, education, credit history, and financial details.
Key Features	- Machine learning-based credit card approval prediction..

	• Real-time prediction using a Flask web application. • User-friendly interface for entering applicant details.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CCPU for model training and deployment	Intel Core i5/i7
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv